Diagnosing Optimization During Fine-Tuning: AdamW vs Muon on SST-2

Experimental setup

I fine-tuned prajjwal1/bert-tiny on the GLUE SST-2 sentiment task using a manual PyTorch training and evaluation loop. I used the official validation split, batch size 32, maximum sequence length 128, 3 epochs, 10% linear warmup, gradient clipping at 1.0, and three seeds (42, 43, 44) on one NVIDIA T4. I gave AdamW and Muon the same data order and training budget for each matched seed. Learning rates were selected with the same small one-seed search using validation loss. Muon was applied only to 2D transformer hidden-weight matrices; embeddings, biases, normalization parameters, and the classifier used AdamW, matching the intended hybrid use of Muon. [1,2]

Metrics and why I used them

I treated final validation loss as the primary performance criterion because it is sensitive to prediction confidence as well as correctness; validation accuracy was the secondary task metric. Training loss shows convergence behavior. Gradient L2 norm helps reveal the scale and stability of the back-propagated signal, while the relative update norm ||Delta theta||/||theta|| measures how strongly the optimizer actually changes the model after transforming the gradient. I also recorded runtime. To probe local flatness, I added layer-relative random parameter perturbations at several magnitudes and measured the resulting change in validation loss. This is a practical sharpness proxy, not a parameterization-invariant definition of flatness. [3]

Main results

Optimizer	Val. loss	Val. acc.	Grad. norm	Rel. update	Runtime (s)
AdamW	0.4621 +/- 0.0115	82.15% +/- 0.24	5.18	5.1e-5	98.0
Muon	0.5066 +/- 0.0111	82.38% +/- 0.75	5.51	4.91e-4	136.1

Interpretation

AdamW performed better by my predefined primary criterion: its mean validation loss was 0.4621 versus 0.5066 for Muon. Muon had slightly higher mean accuracy (82.38% versus 82.15%), but the 0.23 percentage-point difference is small relative to its across-seed variability. The optimizers reached similar training losses, yet Muon produced roughly 9.6x larger relative parameter updates and took about 39% longer. Validation loss increased after the first epoch for both methods, especially for Muon, even while training loss continued to fall; I interpret this as overfitting or worsening calibration rather than evidence that more training would help.

Which solution appears flatter?

The perturbation experiment was not strong enough for a confident flatness ranking. Muon showed a small positive increase in validation loss as perturbation size grew (about +5.6e-4 at rho=0.01), while AdamW showed tiny negative changes (about -3.1e-4 at rho=0.01). The latter means the final AdamW point was not behaving like a clean local minimum of validation loss, or that noise/direction effects were comparable to the measured signal. Therefore I would report the sharpness result as inconclusive rather than claim that AdamW is definitively flatter.

Reliability and what I would change

These conclusions are preliminary: I used one small model, one dataset, three seeds, a narrow learning-rate search, and only three epochs. The sharpness proxy is also scale- and parameterization-sensitive. In a larger study I would save and compare the best-validation checkpoint for each run, use more seeds, tune optimizer-specific learning rates and weight decay more broadly, test additional model sizes/datasets, increase the number of perturbation directions, and add a second curvature measure such as directional finite differences or a top-Hessian-eigenvalue estimate. [3]

References: [1] PyTorch, “Muon” optimizer documentation (accessed Sep. 2026). [2] K. Jordan, “Muon: An optimizer for hidden layers in neural networks,” 2024. [3] L. Dinh et al., “Sharp Minima Can Generalize For Deep Nets,” ICML 2017 / arXiv:1703.04933.